

★ WEEK 13 · PART 2 of 3

Updating in Practice: The Beta-Binomial

Conjugate Priors and the Role of the Prior

The Beta-Binomial · sequential updating · conjugacy · choosing a prior

The abstract update becomes concrete: estimating a probability. The Beta-Binomial makes Bayes' theorem pure arithmetic, shows how belief sharpens with data, and lays bare the role of the prior.

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PART 2 · UPDATING IN PRACTICE

The Beta-Binomial in Full

Estimating a probability with a Beta prior and Binomial data, the conjugate update as counting, how the posterior sharpens with data, and the much-debated role of the prior.

From the Rule to a Real Update

Part 1 gave the theory; Part 2 works an example

1

We have the update rule

Posterior is proportional to likelihood times prior. But abstractly it can feel slippery — we need a concrete case.

2

Estimating a probability is the classic case

How likely is an up-day? A default? The Beta-Binomial answers exactly this, and the update is pure arithmetic.

3

It reveals the key ideas

Conjugacy, sequential updating, and the role of the prior all show up cleanly in this one worked example.

Part 2 in one line: the Beta-Binomial turns Bayes' theorem into counting successes and failures.

Estimating a Probability

The Beta-Binomial: the 'hello world' of Bayes

The simplest useful Bayesian problem: estimate a probability from yes/no data. Put a Beta prior on the probability, observe Binomial data, and the posterior is again a Beta — the update is just arithmetic.

The question

- What is the probability of a success?
- Examples: an up-day, a default, a win.
- We want a distribution over that probability.

The ingredients

- A Beta prior on the probability.
- A Binomial likelihood for the counts.
- A Beta posterior — same family, updated.

The Binomial Likelihood

Counting successes out of trials

Binomial likelihood: $P(k \mid n, \theta) \propto \theta^k (1 - \theta)^{n-k}$

What it says

- Given a probability, how likely is k successes in n ?
- It rewards values near the observed proportion.
- It is the model linking the probability to the data.

Its shape

- Peaked at the sample proportion k over n .
- Sharper as the number of trials grows.
- This is the likelihood in Bayes' theorem.

The Beta Prior

A distribution on a probability in $[0,1]$

$$\text{Beta prior: } \theta \sim \text{Beta}(a, b), \quad p(\theta) \propto \theta^{a-1}(1 - \theta)^{b-1}$$

Why the Beta

- It lives exactly on the interval zero to one.
- So it is a natural prior for a probability.
- Its two parameters a and b flex its shape.

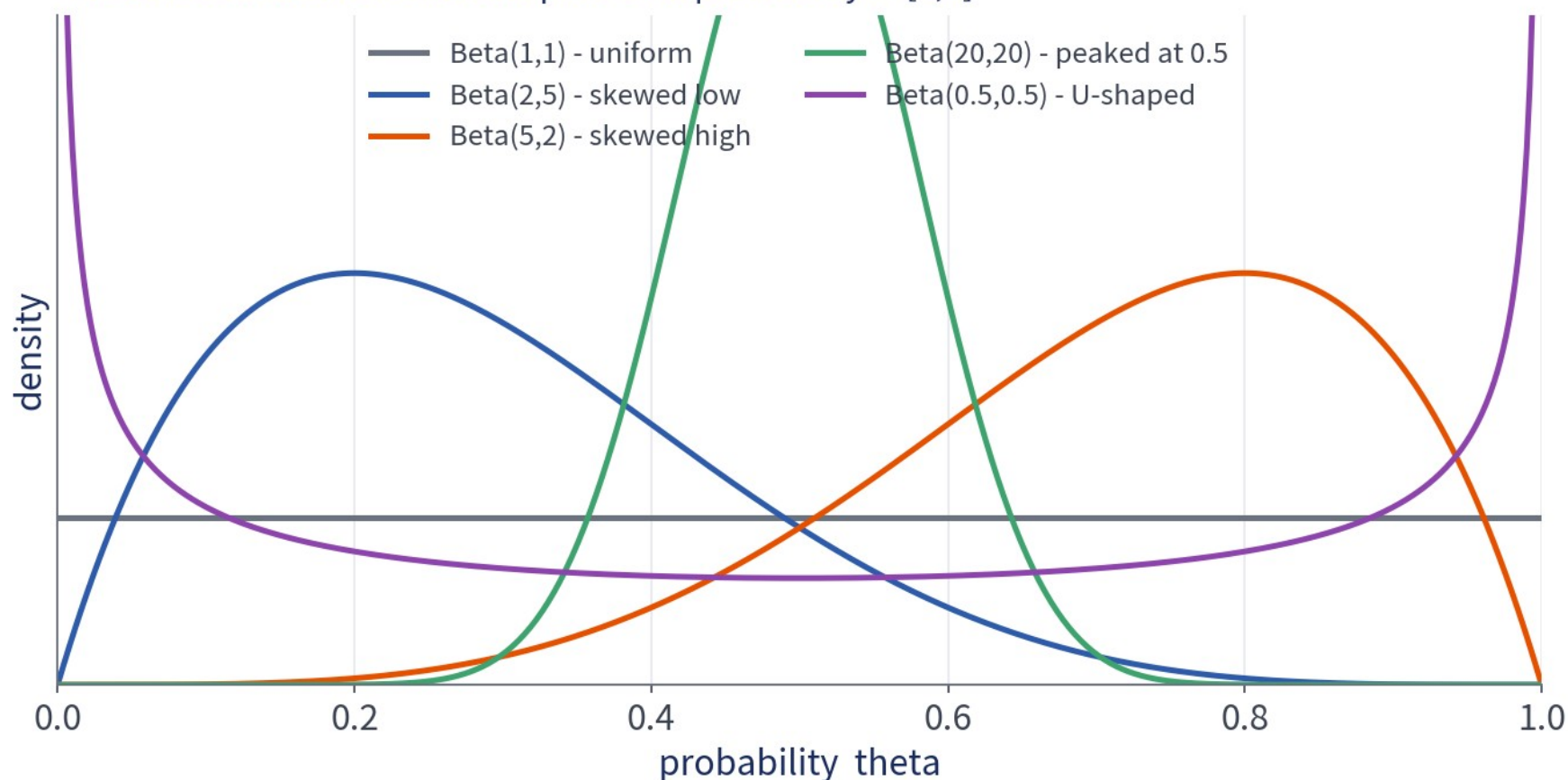
Reading a and b

- They behave like prior successes and failures.
- Larger values make a more confident prior.
- Equal values center it at one half.

The Beta Distribution's Many Shapes

By varying a and b , the Beta can be flat, bell-shaped, skewed, or U-shaped — a flexible family for expressing almost any prior belief about a probability between zero and one.

The Beta distribution: a flexible prior on a probability in $[0,1]$



The Beta-Binomial Update

Add the counts, get a new Beta

$$\text{Beta-Binomial update: } \theta \mid \text{data} \sim \text{Beta}(a + k, b + n - k)$$

The rule

- Add the observed successes k to the prior's a .
- Add the observed failures to the prior's b .
- That's the entire update — no integral needed.

Why it is remarkable

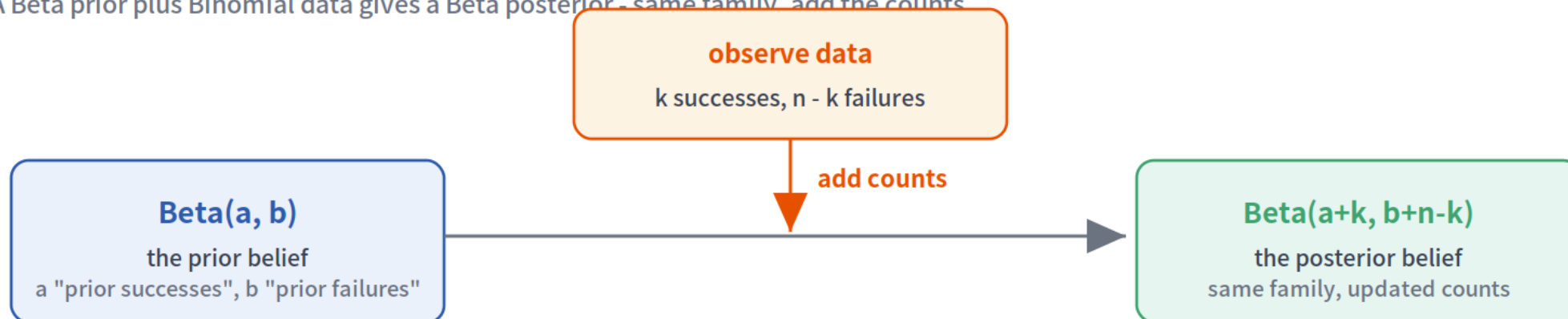
- The posterior is again a Beta distribution.
- So it can serve as the prior for the next batch.
- Bayes' theorem reduces to bookkeeping.

Conjugacy: the Update Is Just Counting

A Beta prior and Binomial data produce a Beta posterior — the same family, with the observed successes and failures simply added to a and b . No integral, no simulation: pure arithmetic.

Conjugacy: the Update Is Just Counting

A Beta prior plus Binomial data gives a Beta posterior — same family, add the counts



Why this is a gift

When the prior and likelihood are CONJUGATE, the posterior is in the same family as the prior. No integral, no simulation - just add the observed successes and failures to the prior's parameters.

a and b act like pseudo-counts: prior evidence, expressed as imaginary past data.

Walking Through the Update

From prior to posterior in one step

Set-up

- Start with a mild prior, say Beta two and two.
- It is centered at one half, weakly held.
- It reflects 'probably balanced, but unsure'.

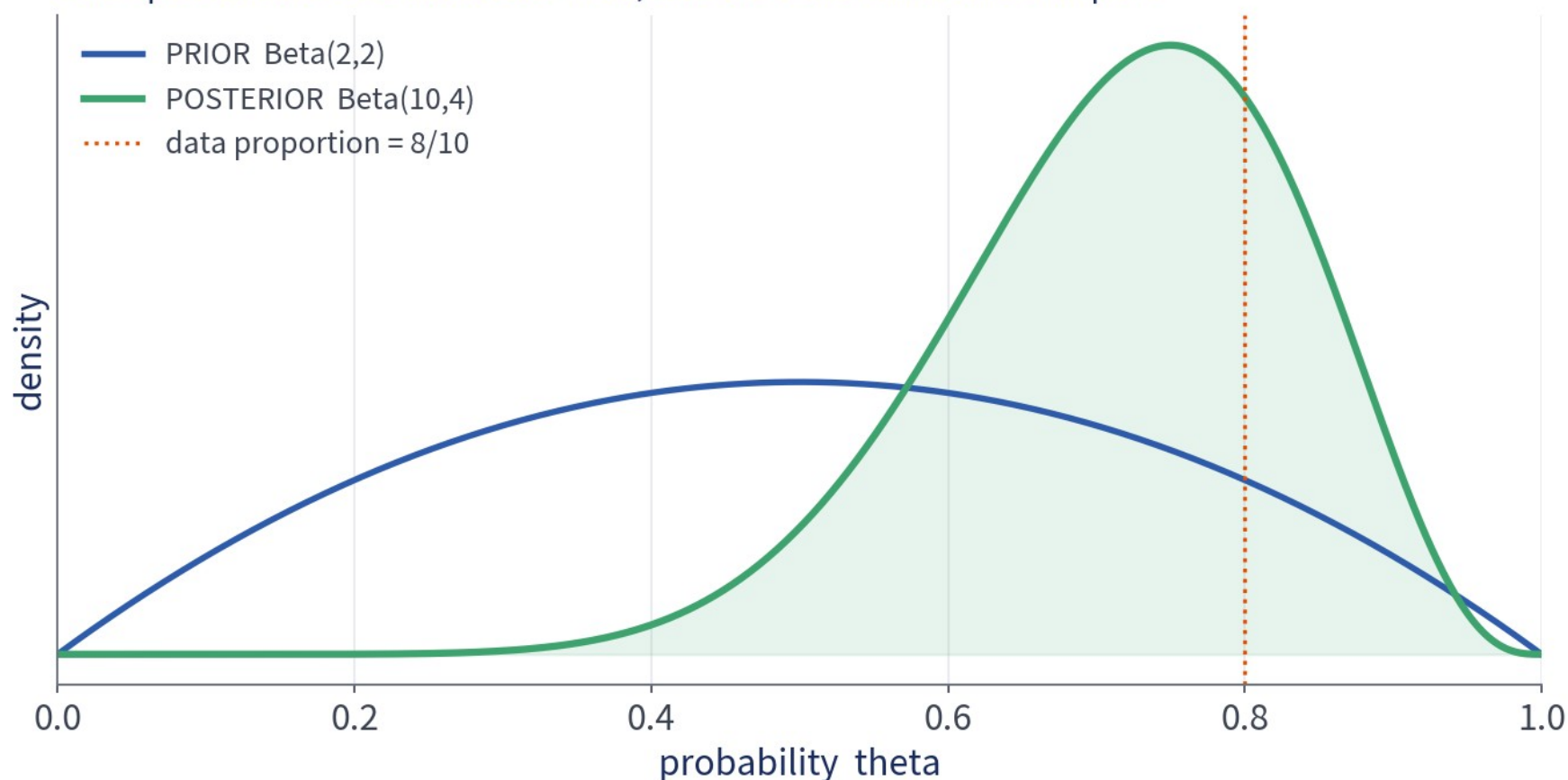
Observe and update

- See eight successes out of ten trials.
- Add the counts: the posterior is Beta ten and four.
- Belief moves toward 0.8 and tightens.

One Update, Start to Finish

A mild prior meets eight successes in ten trials. The posterior shifts toward the observed proportion and narrows — belief has moved decisively, but still carries honest uncertainty.

One update: observe 8 successes in 10, and the belief shifts and sharpens



The Posterior Sharpens With Data

Sequential updating, one Beta at a time

Because the posterior is again a Beta, you can keep updating. Each new observation adds to the counts, and the distribution tightens around the truth. Yesterday's posterior is simply today's prior.

How it accumulates

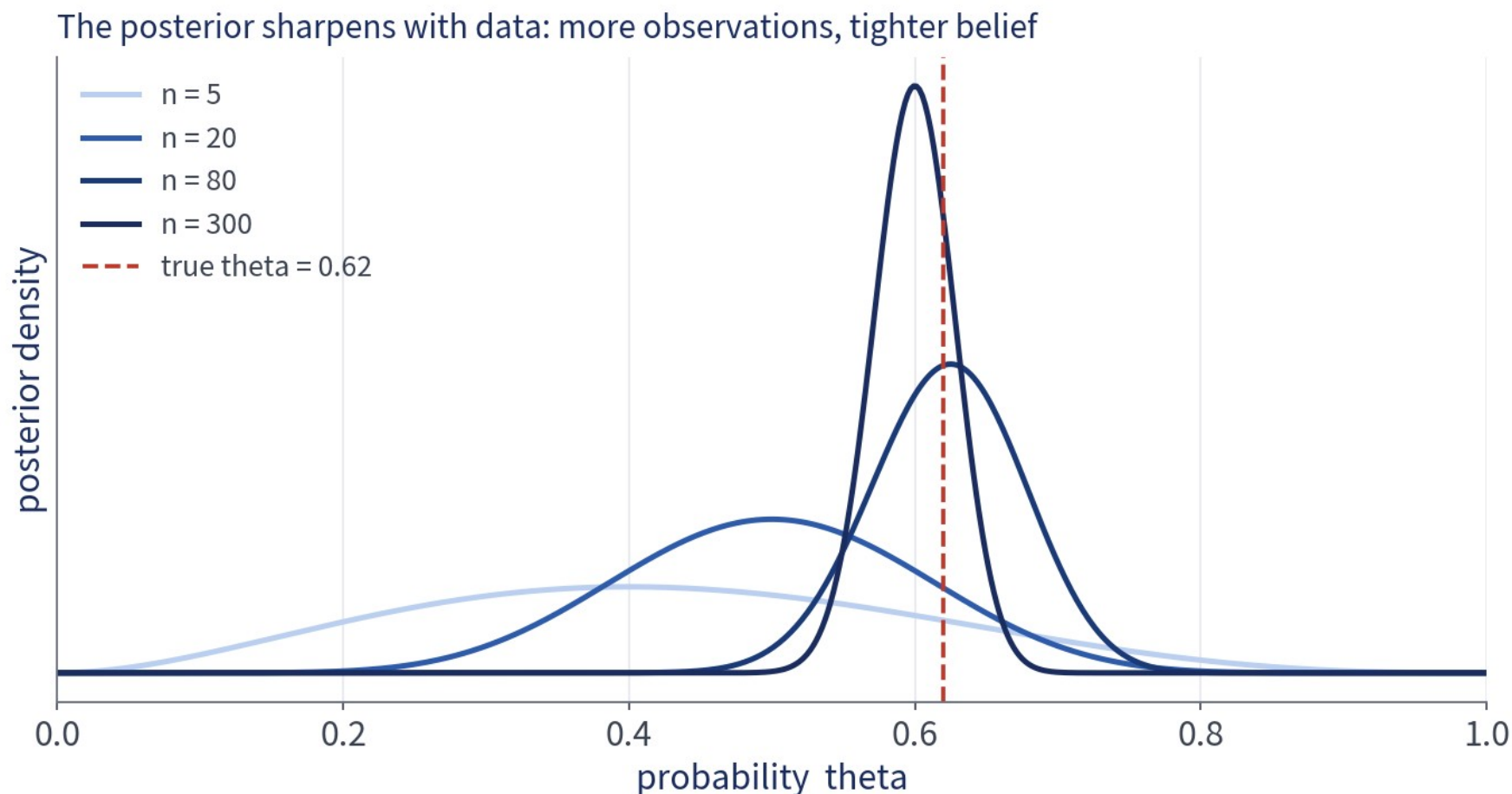
- Feed observations one at a time, or in batches.
- Each just increments a or b .
- The order does not change the final result.

What happens to belief

- The posterior narrows as evidence mounts.
- It concentrates around the true probability.
- Uncertainty shrinks in a principled way.

The Posterior Sharpens With Data

Starting from a flat prior, the posterior tightens around the true probability as observations accumulate. With a handful of points it is broad; with hundreds it is a confident spike.



Conjugate Priors

When the update is just arithmetic

A prior is conjugate to a likelihood when the posterior stays in the same family. Then Bayes' theorem needs no integral — you just update the parameters. The Beta-Binomial is the classic example.

What conjugacy means

- Prior and posterior share the same family.
- The likelihood maps one member to another.
- The update is a parameter change, nothing more.

Why it matters

- No intractable normalizing integral to compute.
- Fast, exact, and easy to reason about.
- It made Bayes usable long before computers.

Common Conjugate Pairs

A small toolbox of exact updates

Prior	Likelihood	Posterior
Beta	Binomial	Beta
Gamma	Poisson	Gamma
Normal (known variance)	Normal	Normal
Dirichlet	Multinomial	Dirichlet

Each pair updates by simple arithmetic — the reason conjugate models are so widely used.

The Posterior Mean as Shrinkage

A blend of prior and data

$$\text{posterior mean} = \frac{a+k}{a+b+n} \quad (\text{a shrinkage of the data toward the prior})$$

Reading it

- The mean blends prior counts with observed counts.
- It sits between the prior mean and the data proportion.
- It shrinks the raw estimate toward the prior.

Why shrinkage helps

- With little data, it leans on the prior — more stable.
- With much data, the observed counts dominate.
- It avoids wild estimates from tiny samples.

The Update as Counting

Priors as pseudo-counts

Prior as imaginary data

- The prior's a and b act like past successes and failures.
- A Beta two and two is like having seen two of each.
- The prior is 'pretend evidence' you start with.

Data as more counts

- Real observations add to those pseudo-counts.
- The posterior mean is just total successes over total trials.
- Big priors resist data; small priors yield to it.

The Role of the Prior

The most debated — and most useful — ingredient

The prior is where Bayesians are challenged: isn't it subjective? Yes — but it is an honest, explicit input, and a powerful one. It injects knowledge, stabilizes small samples, and fades as data accumulates.

The criticism

- The prior is a choice, and choices can be arguable.
- Different analysts might pick different priors.
- So conclusions can depend on that choice.

The defense

- The prior is explicit and open to scrutiny.
- It encodes genuine domain knowledge.
- With enough data, its influence disappears.

Informative vs Weak Priors

How much to let belief speak

Informative prior

- Encodes real, specific prior knowledge.
- Narrow — it resists being moved by little data.
- Powerful when you truly know something.

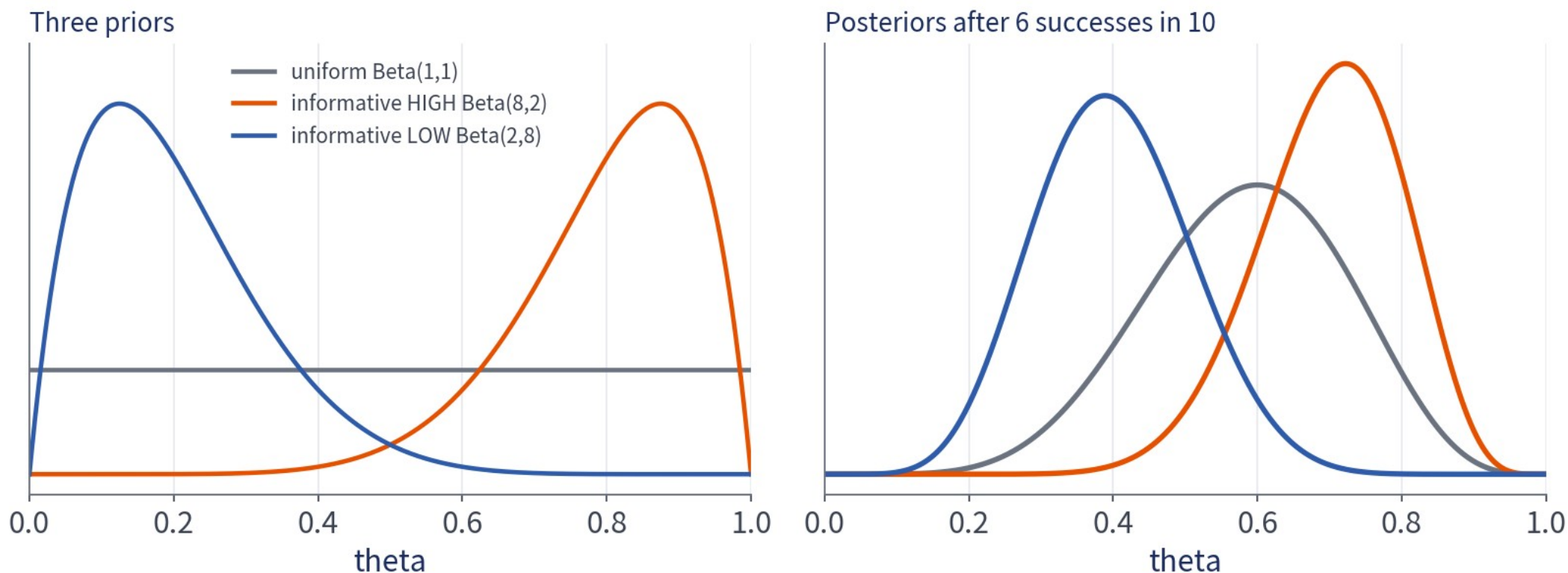
Weak (diffuse) prior

- Deliberately vague — lets the data lead.
- Wide and nearly flat over the range.
- A safe default when you know little.

Same Data, Different Priors

With only ten observations, three different priors yield three different posteriors. The prior still matters when data is limited — which is exactly when its stabilizing knowledge is most valuable.

Same data, different priors: the prior still tilts the posterior when data is limited



Priors as Regularization

Stabilizing small-sample estimates

The problem it solves

- Tiny samples give wild, unstable estimates.
- Zero successes doesn't mean zero probability.
- Raw proportions can be nonsensical at the extremes.

How the prior helps

- It pulls estimates toward a sensible default.
- This shrinkage is a form of regularization.
- It trades a little bias for much less variance.

When the Prior Washes Out

And how to choose one responsibly

The washout

- As data grows, the likelihood dominates.
- The posterior converges regardless of a reasonable prior.
- Priors matter most exactly when data is scarce.

Choosing responsibly

- Prefer weakly-informative priors as a default.
- Run a sensitivity check across plausible priors.
- Report how much the prior actually mattered.

Beta-Binomial Sequential Updating

Direct Claude Code to watch belief sharpen

GOAL Estimate a probability sequentially with a Beta-Binomial and see the posterior tighten as data arrives.

[INPUT] Prompt · Claude Code

Write and run `beta_binomial.py` that:

1. sets a true probability and simulates a stream of Bernoulli (yes/no) outcomes;
2. starts from a $\text{Beta}(a, b)$ prior and updates it observation by observation (add to a or b);
3. plots the posterior after 5, 20, 80, and 300 observations, with the true value marked;
4. also compares two different priors on the same data to show the prior washing out.

Confirm the posterior concentrates on the truth.

How to work with it

- A success adds one to a ; a failure to b .
- The posterior narrows as n grows.
- Different priors converge with enough data.
- The posterior mean is a shrinkage estimate.

Deliverable: `beta_binomial.py` + sequential posteriors and a prior-sensitivity comparison.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

After k successes and $n-k$ failures, a $\text{Beta}(a,b)$ prior becomes what?

$\text{Beta}(a+k, b+n-k)$

Q2

What does 'conjugate' mean for a prior?

Posterior stays in the same family

Q3

When does the choice of prior matter most?

When data is scarce

All three come from the update rule, conjugacy, and the prior's fading influence.

Three Misconceptions to Avoid

Where Beta-Binomial intuition slips

Common myths

- "A conjugate prior is chosen because it's true."
- "The prior always biases the answer."
- "The order of the data changes the posterior."

The reality

- Conjugacy is chosen for convenience, not truth.
- With enough data the prior washes out entirely.
- For this model, the posterior is order-independent.

Three Things to Carry Forward

Part 2 in one screen

1

The Beta-Binomial is Bayes made concrete

A Beta prior and Binomial data give a Beta posterior. The update is pure arithmetic: add the observed successes and failures to the prior's parameters.

2

Conjugacy makes updating trivial

When prior and likelihood are conjugate, the posterior stays in the same family — no integral needed. The prior's parameters act as pseudo-counts, and the posterior mean is a shrinkage estimate.

3

The prior is honest and fades

It injects knowledge and stabilizes small samples, and it matters most when data is scarce. As evidence mounts, the likelihood takes over and the prior washes out.

Next — Part 3: credible intervals, the posterior predictive, and why Bayes suits finance.